

📘 NOTE 5 — SQL COMMANDS AND QUERIES

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Subject: DBMS
Topic: SQL Commands and Queries
Level: BCA Semester 5

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1. Introduction to SQL

SQL stands for Structured Query Language.

SQL is a standard language used to communicate with relational databases. It allows users to create tables, insert data, retrieve records, update information, and delete records.

SQL is commonly used with database systems such as **MySQL, Oracle, PostgreSQL, and SQL Server**.

2. Types of SQL Commands

SQL commands are commonly divided into the following categories:

DDL — Data Definition Language

Used to define the structure of database objects.

Examples:

- `CREATE`
- `ALTER`
- `DROP`
- `TRUNCATE`

DML — Data Manipulation Language

Used to add and modify data.

Examples:

- `INSERT`
- `UPDATE`
- `DELETE`

DQL — Data Query Language

Used to retrieve data.

Example:

- `SELECT`

DCL — Data Control Language

Used to control database permissions.

Examples:

- `GRANT`
- `REVOKE`

TCL — Transaction Control Language

Used to manage transactions.

Examples:

- `COMMIT`
- `ROLLBACK`

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3. CREATE TABLE

The `CREATE TABLE` command is used to create a new table.

Example:

- CREATE TABLE Students (
- student_id INT PRIMARY KEY,
- name VARCHAR(100),
- email VARCHAR(100),
- semester INT

);

This creates a `Students` table with four columns.

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4. INSERT Command

The `INSERT` command is used to add records to a table.

- INSERT INTO Students

- (student_id, name, email, semester)
- VALUES

(101, 'Aarav Mehta', 'aarav@example.com', 5);

Another record can be added:

- INSERT INTO Students
- VALUES

(102, 'Rohan Shah', 'rohan@example.com', 5);

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5. SELECT Command

The `SELECT` command retrieves data from a table.

Display all records

SELECT * FROM Students;

Display selected columns

- SELECT name, email

FROM Students;

The `*` symbol means all columns.

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6. WHERE Clause

The `WHERE` clause is used to filter records according to a condition.

- SELECT *
- FROM Students

WHERE semester = 5;

This query displays only students whose semester is 5.

Another example:

- SELECT *
- FROM Students

WHERE student_id = 101;

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7. UPDATE Command

The `UPDATE` command is used to modify existing records.

- UPDATE Students
- SET semester = 6

WHERE student_id = 101;

The `WHERE` condition is important because it specifies which record should be updated.

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8. DELETE Command

The `DELETE` command is used to remove records.

- DELETE FROM Students

WHERE student_id = 102;

This removes the student whose ID is `102`.

Important: Always use an appropriate `WHERE` condition when deleting specific records.

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9. ORDER BY

`ORDER BY` is used to sort query results.

Ascending order

- SELECT *
- FROM Students

ORDER BY name ASC;

Descending order

- SELECT *
- FROM Students

ORDER BY name DESC;

`ASC` means ascending and `DESC` means descending.

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10. LIKE Operator

The `LIKE` operator is used to search for a specific pattern.

- SELECT *
- FROM Students

WHERE name LIKE 'A%';

This finds students whose names start with **A**.

Here, `%` represents zero or more characters.

Example:

WHERE email LIKE '%example.com';

This searches for emails ending with `example.com`.

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11. Aggregate Functions

Aggregate functions perform calculations on multiple records.

Common aggregate functions are:

- `COUNT()` — counts records
- `SUM()` — calculates total
- `AVG()` — calculates average
- `MAX()` — finds maximum value
- `MIN()` — finds minimum value

Example:

- SELECT COUNT(*) AS TotalStudents

  FROM Students;

This returns the total number of students.

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12. GROUP BY

`GROUP BY` is used to group records with similar values.

Suppose an `Assignments` table contains different subjects.

- SELECT subject, COUNT(*) AS total
- FROM Assignments

  GROUP BY subject;

This can show the number of assignments available for each subject.

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13. SQL JOIN

A `JOIN` combines data from two or more related tables.

For example:

- Students
- ↓

Submissions

A student can have multiple assignment submissions.

INNER JOIN Example

- SELECT Students.name, Submissions.assignment_id
- FROM Students
- INNER JOIN Submissions

ON Students.student_id = Submissions.student_id;

This retrieves student information along with their submissions.

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14. SQL in StudySphere

SQL queries can be used in the PHP/MySQL version of StudySphere.

For example:

Register Student

- INSERT INTO users
- (name, email, password)
- VALUES

('Aarav Mehta', 'aarav@example.com', 'password');

Display Notes

- SELECT *

FROM notes;

Find Notes by Subject

- SELECT *
- FROM notes

WHERE subject = 'DBMS';

Display Assignments

- SELECT *

FROM assignments;

Update Grade

- UPDATE submissions
- SET grade = 85

WHERE submission_id = 1;

These operations help StudySphere perform its main database functions.

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15. CRUD Operations

CRUD represents the four basic database operations:

C — Create
Add new data.

INSERT INTO Students VALUES (...);

R — Read
Retrieve data.

SELECT * FROM Students;

U — Update
Modify existing data.

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UPDATE Students SET ...;
```

D — Delete
Remove data.

```
DELETE FROM Students WHERE ...;
```

CRUD is one of the most important concepts in database-driven applications.
